Maintenance Access in Hydropower Plants: Permanent Platforms and Scaffolding

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Research Question

For turbine maintenance which has to be done over and over again, what are the practical differences between using a permanent modular platform and setting up temporary scaffolding each time? This review looks at the fall hazards which come with each system, the amount of time which is needed to get access set up before the work can start, and the inspection and maintenance work which comes with each system.

Summary

From the sources which were reviewed, the main benefit of a permanent platform does not seem to come from the inspection itself. It comes from the work which has to happen before and after each inspection. The scaffold has to be built, checked, and then taken down again, and this is repeated every time access is needed. At the same time, the workers who are doing this are exposed to fall hazards while the scaffold is still incomplete. The sources which were found suggest that a permanent arrangement would be safer and would also take less time to get onto. However, no comparison was found where the hours saved or the incidents avoided were actually measured, and there was also very little information on what is needed to maintain a permanent platform over its life. Because of this, the case for a permanent platform at the moment is based on the reduction in repeated scaffold work rather than on any number which can be quoted.

1. Background

In many cases, inspecting the underside of a turbine runner is more of an access problem than an inspection problem. The work has to be done inside the draft tube, where there is not much room to move around. The usual approach is to build scaffolding inside the draft tube elbow so that the inspection areas can then be reached. After the work is finished, the scaffold is then removed. If access is needed again during a later outage, the same arrangement then has to be built again from the beginning.

Hatch (2026) published a different approach in January 2026 [1]. A modular access platform was developed for SCHM's McCormick Hydropower Plant in Québec. The project started in 2019 after a safety concern was found underneath the turbine runner during an inspection. From the information which Hatch provided, the finished system serves five turbine-generator units, works with three different turbine types, and removes the need for conventional scaffolding.

The reason this project was chosen is because turbine maintenance is repeated. A temporary solution may be fine when access is only needed once in a while. It becomes a lot less attractive when essentially the same working area has to be set up again every time the equipment is inspected. At the same time, this does not automatically mean that the permanent installation is the better option. It still has to be designed for the site, inspected, and also maintained. Because of this, what needs to be looked at is whether the amount of scaffold work which is avoided is actually worth more than the extra responsibilities which then come with the permanent platform.

2. Sources and Scope

Six publicly available sources were reviewed. These include Ontario safety legislation, provincial working-at-heights guidance, a University of Toronto scaffold standard, one peer-reviewed study, and project information which was published by Hatch and by ZARGES. The legislation and safety documents were used to find the requirements which apply to scaffolding and fall protection. The Hatch and ZARGES publications were then

used to look at the alternatives to conventional access which are currently being used in hydropower stations.

There is a limitation with using company publications for this kind of comparison. The companies explain what their own systems are supposed to do. At the same time, the operating records which would be needed to actually check those benefits are not included, so the claims cannot be confirmed from the publications on their own. Ontario Regulation 213/91 was also used as the reference regulation. This is because it has specific requirements in it for scaffolding and also for fall protection. It is not being presented as the regulation which governs McCormick, since McCormick is located in Québec. This review also did not try to verify the structural capacity of either access system or estimate the cost of building a permanent platform. Both of these would have to be done separately.

Table 1. Information available from the reviewed sources

Reference	Relevant information	What remains unclear
Hatch [1]	Describes the McCormick platform and the problems which come with repeated scaffolding	Labour hours, incident data and maintenance requirements
Ontario guidance [2]	Explains mandatory working-at-heights training	Does not compare access systems
O. Reg. 213/91 [3]	Gives the requirements which apply to fall protection, guardrails and scaffolding	Does not say anything about replacing scaffolding with a permanent platform
UofT EHS [4]	Gives scaffold erection and inspection guidance	Does not address permanent turbine platforms
Robson et al. [5]	Reports changes in fall-related injury claims after the Ontario training requirement	Does not measure the effect of permanent platforms
ZARGES [6]	Describes quick setup of portable access equipment at a hydropower station	Not directly comparable to McCormick's permanent system

3. Discussion

3.1 Fall Hazards

When scaffolding is being considered, it is easy to only think about the finished platform which the workers end up standing on. However, someone then has to build it first. During erection, workers may be at height before the whole scaffold and its protective systems are complete. The same problem then comes up again during dismantling. These activities are separate from the turbine inspection, but they still expose workers to hazards just so that the inspection can take place.

There are a few parts of Ontario Regulation 213/91 which then apply to this kind of work [3]. In Section 26, fall protection is required where a worker could fall more than three metres. In Section 26.3, guardrails are required along scaffold and platform edges where the distance which the worker could fall is 2.4 metres or more. Section 26.1 then also states that guardrail systems should be used ahead of the personal fall protection methods. This difference matters here, since a worker who is standing on a finished and properly guarded platform is in a very different situation from a worker who is assembling access equipment while some of that protection is not in place yet. The University of Toronto scaffold standard also covers protection for workers who are doing erection work at height for the same reason [4]. In addition, the mandatory working-at-heights training in Ontario applies to workers who use fall protection devices such as travel restraint and fall arrest systems, and guardrails are not included in that list [2], which is consistent with the preference for guardrail systems which is given in Section 26.1 [3].

Taking this into consideration, a permanent platform with proper guarding could avoid the repeated exposure which happens during scaffold assembly and removal. This would be a meaningful advantage for a station

which needs access during several outages. It would not make the work free of fall hazards though. The platform would still need to provide safe access, proper edge protection, and also protection for any activity which takes a worker outside of the guarded area.

Robson et al. (2024) give some useful background on fall prevention [5]. Their study examined the mandatory working-at-heights training in Ontario. From their results, the lost-time claim rate for targeted fall injuries was 19% lower in 2017–19 than in 2012–14, compared to a 6% reduction in other provinces over the same period. These percentages should not be taken as the expected improvement from installing a permanent platform, since the study looked at training and not at alternative access structures. The more defensible conclusion is a narrower one. Avoiding scaffold erection and dismantling removes two recurring activities where fall exposure can happen. The actual reduction in risk would then have to be worked out for the specific station and work procedure.

3.2 Time Required to Establish Access

Time becomes a more important consideration when the same access equipment has to be installed again and again. For a scaffold arrangement, the work does not start when the inspection crew steps onto the platform. The components first have to be brought into position, assembled, and then inspected. Once the maintenance is finished, the structure then has to be dismantled and removed as well. At a generating station this work also has to fit within the planned outage. For this reason, the time which is spent setting up access is worth looking at separately from the inspection itself.

Hatch (2026) identifies scaffold installation as a time-consuming part of the conventional method [1]. However, the article does not give the crew size, the number of hours which are normally needed, or the effect on the outage schedule. Because of this, the size of the saving at McCormick cannot be worked out from it.

ZARGES gives another example from the Töging am Inn hydropower station in Germany [6]. The station has fourteen turbines, and one unit goes through a major overhaul each year over approximately three months. Portable work platforms were used in place of scaffolding for components up to four metres high, and ZARGES states that the equipment can be ready to use in less than a minute. This number needs some context. The ZARGES equipment is a portable access system. It is not the same as the permanent modular arrangement which was developed for McCormick, and so the setup time cannot be carried over from one system to the other. Both projects show why alternatives to scaffolding are being considered, but neither one gives enough data to calculate the time which would actually be saved.

A useful comparison would start from the existing maintenance procedure. Some examples of what would be needed are the number of workers who are assigned to scaffold erection, how long the installation takes, how often it is repeated, and whether the scaffold removal affects when the generating unit can go back into service. Once these figures are available, the setup and removal time for the permanent system can then be compared against the existing approach. Without them, a numerical claim about improved efficiency would be difficult to support.

3.3 Inspection and Maintenance

The inspection requirements are where the comparison becomes less straightforward. Temporary scaffolding has to be checked before workers use it. Ontario Regulation 213/91 covers scaffold design and inspection in Section 130 and the supervision of erection, alteration, and dismantling in Section 131 [3]. Under Section 130(1), a scaffold exceeding 15 m above its base, or 10 m for a tube-and-clamp scaffold, is also subject to additional design requirements. The University of Toronto standard also requires scaffold inspections after erection and after relocation [4]. Since the scaffold is assembled again and again, many of these activities then have to be repeated during each maintenance outage.

A permanent platform would not have the same recurring erection process. However, it would bring long-term inspection responsibilities of its own. This is particularly relevant inside a draft tube. The moisture and the operating environment mean that corrosion protection has to be considered, and the connections, supporting members, and also the guardrails would all have to stay in acceptable condition over the service life of the platform. The available publications do not explain how the McCormick platform is maintained. The

inspection interval, intended design life, protective coating system, and whether inspection responsibility sits with the station owner or with another party could not be found. Because of this, a fair comparison of the total maintenance effort is not possible from the information which is available. It would also be premature to conclude that the permanent platform needs less inspection just because it does not have to be erected for every outage. Its inspections may be less frequent, but their scope and complexity still have to be determined.

4. Limitations of the Review

Several points would need to be resolved before this research could be used to support an actual project decision.

The first is the applicable legislation. McCormick is located in Québec, and the relevant Québec requirements were not reviewed. Ontario Regulation 213/91 is a useful reference for discussing scaffolding, but it cannot be assumed to apply to the McCormick installation. There is also a further distinction for an Ontario station. Depending on the type of work, the applicable requirements may fall under O. Reg. 213/91 for construction projects [3] or under Reg. 851 for industrial establishments [7]. This determination would have to be made for the actual maintenance activity which is being carried out.

The second limitation is the lack of measured operating data. None of the sources gives a complete comparison of scaffold labour hours, permanent platform setup time, project costs, or the incidents which are associated with each arrangement. In addition, there is a concern about the independence of the information. Hatch and ZARGES are both describing solutions which are connected to their own businesses. Their publications help to explain the applications, but independent records would give much stronger support for any claim about performance.

Finally, the provisions of O. Reg. 213/91 were reviewed using the IHSA's reproduction of the regulation. The cited sections and the current wording should be confirmed through Ontario's e-Laws before they are relied on in a formal engineering assessment.

5. Conclusion

The McCormick project shows how a permanent modular platform can be used to deal with repeated maintenance access inside a hydropower station. At the same time, it also brings up the question of how the access equipment gets chosen in the first place. The scaffolding itself may be fine for the work which is being done on it. The problem is that every time it is installed again, more time is used up and the workers are also exposed to the hazards which come with putting it up and taking it down. In a station where the inspections happen often, this repeated work then ends up being a large part of the maintenance which is being done.

Of the three factors which were considered, worker safety gives the clearest reason to look into a permanent platform. Reducing the need to build and remove temporary scaffolding should reduce the exposure to those particular activities, although the size of the improvement was not measured in any of the sources which were reviewed. The time which could be saved is also worth looking into. The problem is that there is not enough published data at the moment to work out what that saving would actually be. Inspection and maintenance are also still not settled. A permanent platform avoids the repeated scaffold erection, but it then has to be kept structurally sound and serviceable for its whole intended life.

Taking all of this into consideration, selecting one system over the other based only on the available publications would not be recommended. Before doing this, a record of the current scaffold labour requirements, the frequency of maintenance access, and the expected service life and inspection needs of the proposed platform would be needed. These details would turn the comparison from a general argument in favour of permanent access into something which could actually support an engineering decision.

6. Questions for Senior Engineering Review

1. When turbine maintenance has to be repeated, is getting rid of the scaffold erection and dismantling enough on its own to justify a permanent platform from a safety point of view, or would measured labour hours and outage data normally be needed first before that decision is made?
2. Which legislation and which site requirements would then apply to the access system? In particular, how is it normally decided whether the work counts as construction or as ongoing industrial maintenance at a station like this one?
3. For a permanent platform which has been installed inside a draft tube, who is normally responsible for inspecting it over its service life (the station owner, the designer, or someone else), and how often and to what level of detail would that inspection then be expected to be done?

References

[1] Hatch Ltd., “Setting a new standard: Hatch’s modular platform enhances safety and efficiency in hydropower,” news release, Jan. 15, 2026. hatch.com/About-Us/News-And-Media/2026/01/Setting-a-new-standard. Accessed Oct. 18, 2026.

[2] Government of Ontario, “Training for working at heights,” updated Apr. 21, 2026. ontario.ca/page/training-working-heights. Accessed Oct. 18, 2026.

[3] O. Reg. 213/91, Construction Projects, ss. 26, 26.1, 26.3, 130, 131, as reproduced in IHSA OHS Guide, ohsguide.ihsa.ca (Guardrails; Scaffolds – Erection; Scaffolds – Inspection). Accessed Oct. 18, 2026.

[4] University of Toronto EHS, “Working at Elevated Places – Scaffolds Standard,” Aug. 2022. Accessed Oct. 18, 2026.

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[6] ZARGES GmbH, “Safety turbine maintenance using ZARGES work platforms,” undated. zarges.com. Accessed Oct. 18, 2026.

[7] R.R.O. 1990, Reg. 851, Industrial Establishments, under the Occupational Health and Safety Act. Ontario e-Laws. Accessed Oct. 18, 2026.